Edouard Murat

Gameplay Programmer & Team Lead

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French: native • English: fluent

PROFILE

Gameplay programmer and team lead with 20 years of software experience, including five years of production C#. Strongest at getting software from plan to release: planning the work, building the core systems, and carrying the product through review and launch. Led a six-person Godot team building curriculum-aligned educational games for Android. Specializes in gameplay systems, AI behavior, procedural generation, shader programming, and reusable game frameworks. Five games live on Steam in a free public demo, with deployment pipelines for Google Play and the Apple Store. The common thread is software that teaches: a psychology degree, children's literacy and numeracy games at Sigma Digital, and an independent educational games platform.

TECHNICAL SKILLS

Game Engines	Godot, Unity
Languages	GDScript, C#, Python, Java, HTML/CSS/JS, SQL
Shaders & Graphics	Shader scripting (Godot, Unity)
Gameplay Systems	Input (keyboard & mobile), AI behaviors, procedural generation, physics-based mechanics, state machines, combat systems
Tools & Frameworks	Git, GitHub, Blender, VS Code, Django, SQLite, Audacity, Linux
Other	Agile workflows, UI integration, performance optimization

EXPERIENCE

Lead Gameplay Programmer (Godot) — Sigma Digital, South Africa (remote)

Sept 2025 – Aug 2026

Led a six-person Godot team building curriculum-aligned literacy and numeracy games delivered on Android tablets to first- and second-grade children in Africa.

- Hired the programming team and led it as technical lead, owning architecture, code review, and sprint planning
- Designed the platform's UI framework and an MVVM + FSM architecture adopted across all activities, decoupling game logic from presentation, enabling per-activity unit testing, and making curriculum content swappable without rebuilding core mechanics
- Delivered six of a planned 200 mini-games and supported two rounds of playtesting with the target age group
- Built for low-cost Android tablets, with gameplay mapped to a first- and second-grade reading, writing and math curriculum
- Worked directly with educators and instructional designers to translate curriculum goals into game mechanics, bridging educational theory and interactive design across each sprint
- Provisioned GPU-enabled AWS EC2 instances to run automated Godot UI tests, enabling headless rendering for CI without local hardware

Game Developer — Playworks North, Montreal

2022 – Present

Founder and lead developer of an educational gaming platform blending custom-built quizzes with gameplay across farming, hack & slash, platformer, puzzle, and sports genres.

- Shipped five games in Unity and Godot, live on Steam as a free public demo since September 2026, integrating real-time quizzes into diverse gameplay loops
- Designed and implemented gameplay systems: player control, AI, procedural generation, level logic, scoring, and feedback
- Implemented unified input systems supporting both keyboard and mobile touch controls
- Created custom shaders for visual storytelling and UX (X-ray vision, mini-cameras, 3D text, underwater effects)
- Built a full-stack Django backend: authentication, data storage, in-game unlockables, and content delivery
- Built and maintained deployment pipelines for Steam, Google Play, and the Apple Store; took the Steam demo through Valve review to public release

Gameplay Programmer (Godot) — Shattered Paradise, One More Turn Studios (remote)

Jul 2025 – Dec 2025

Sole programmer on a turn-based grid-based dungeon crawler roguelike (Dante's Inferno theme, procedurally generated dungeons). The creative director defined the vision; owned all technical work end-to-end.

- Wrote all Godot gameplay code, shaders, the game website and backend from scratch, prioritizing and building features independently

Gameplay Programmer (Godot) — Pursuit of Life, Dreamboat (remote)

Jun 2025 – Aug 2025

Shader-focused contract on a pixel-art urban life-sim.

- Built a dynamic rain system with customizable impact zones and per-region wetness tracking
- Implemented real-time puddle reflections for sprites, NPCs, and swaying foliage
- Developed day/night lighting driven by a global time-of-day system

Web Developer and Graphic Designer — HeadPause, Montreal 2018 – 2020

- Designed, built and maintained the organization's website and its visual assets (animations, posters, logo); ran its email campaigns and paid advertising (Facebook Ads, Mailchimp, AdWords)

Java and Web Developer — BMC Software, Montreal 2013 – 2014

- Built a new adaptive client-facing site in Java and AngularJS, alongside parallel remote teams
- Code review, TDD and quality control; shared management of the development team in SCRUM (sprint planning) and interviewed candidates for web developer positions

C# Developer and Analyst — Fujitsu Consulting, Montreal 2008 – 2013

- Five years of production enterprise C#: a payment solution for HydroSherbrooke (utility) and the daycare management system for the Ministère de la Famille du Québec (government)
- Gathered requirements directly from clients, resolved blocking production incidents and delivered across multi-year projects
- Trained incoming developers over a three-year engagement and acted as the technical reference for the codebase

SELECTED GAME PROJECTS

The five EdNoKa titles in the Steam demo; full list of playable games at edouardmurat.com

Base of 30 (Godot 2025, rebuilt in Unity 6) Rogue hack & slash: procedural room layouts and navigation, behavior-tree enemies with distinct attack patterns, combat and progression

Aerial Ambush (Godot 2025, rebuilt in Unity 6) Skill-based platformer: high-speed movement with precise collision, homing-missile AI with physics-based steering

Agent Ka (Godot, 2024) Waldo-style search in a generated city: procedural city and crowd simulation with density logic; X-ray, mini-camera and slow-motion powers as shaders

Veggies Island (Godot, 2024) Farming on a procedural island: planting-to-harvest growth loop and inventory

Pico Golf (Godot, 2023) 3D mini golf: physics-based terrain, water and caustics shaders, animated fish

Earlier titles include Tomb (Godot Wild Jam #42, built in a week), Left 4 Dead 2D, Meditation Card Game, Fruit, Archer and Meditation World VR.

EDUCATION

B.A. Psychology — Concordia University, Montreal 2015 – 2017

B.Eng. Computer Engineering — University of Sherbrooke 2005 – 2009

College Diploma, DEC in Sciences — Stanislas College, Montreal 2003 – 2005

CERTIFICATIONS

Certified Associate Python Programmer — Python Institute 2024

Oracle Professional Certification, Java SE 7 Programmer 2013

Oracle Associate Certification, Java SE 7 Programmer 2012

References available upon request.